README(How to run the code) – Value Iteration in Grid World

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Project Description

This project implements the **Value Iteration algorithm** to solve a Grid World Markov Decision Process (MDP).

The program:

- Computes the **optimal state values V(s)**
- Extracts the **optimal policy π(s)**
- Displays the results using a GUI
- Saves the final policy as images

How to Run the Program

1. Open terminal or command prompt
2. Navigate to the project folder
3. Run the program:

python MDP.py

4. Enter the grid file name when prompted: for example:

gridworld_easy-1.txt

Output

The program produces:

- State values for each grid cell
- Optimal policy (arrows)
- Number of iterations for convergence

- Saved images for each discount factor

Discount Factors Used

The following values of γ (gamma) were used:

- γ = 0.90
- γ = 0.95
- γ = 0.99

Terminal Output (Easy Grid)

```
PS C:\Users\banja\OneDrive\Desktop\Project1_CSCE4201\gridworld-value-iteration> python MDP.py
Enter the gridworld spec file name:
gridworld_easy-1.txt
Gamma = 0.9 converged in 7 iterations
Gamma: 0.9, Iterations: 7
Saved: gridworld_easy-1_gamma_0_9.png
Gamma = 0.95 converged in 7 iterations
Gamma: 0.95, Iterations: 7
Saved: gridworld_easy-1_gamma_0_95.png
Gamma = 0.99 converged in 7 iterations
Gamma: 0.99, Iterations: 7
Saved: gridworld_easy-1_gamma_0_99.png
```

Terminal Output (Medium Grid)

```
PS C:\Users\banja\OneDrive\Desktop\Project1_CSCE4201\gridworld-value-iteration> python MDP.py
Enter the gridworld spec file name:
gridworld_medium-1.txt
Gamma = 0.9 converged in 28 iterations
Gamma: 0.9, Iterations: 28
Saved: gridworld_medium-1_gamma_0_9.png
Gamma = 0.95 converged in 31 iterations
Gamma: 0.95, Iterations: 31
Saved: gridworld_medium-1_gamma_0_95.png
Gamma = 0.99 converged in 35 iterations
Gamma: 0.99, Iterations: 35
Saved: gridworld_medium-1_gamma_0_99.png
```

Terminal Output (Hard Grid)

```
PS C:\Users\banja\OneDrive\Desktop\Project1_CSCE4201\gridworld-value-iteration> python MDP.py
Enter the gridworld spec file name:
gridworld_hard-1.txt
Gamma = 0.9 converged in 29 iterations
Gamma: 0.9, Iterations: 29
Saved: gridworld_hard-1_gamma_0_9.png
Gamma = 0.95 converged in 33 iterations
Gamma: 0.95, Iterations: 33
Saved: gridworld_hard-1_gamma_0_95.png
Gamma = 0.99 converged in 40 iterations
Gamma: 0.99, Iterations: 40
Saved: gridworld_hard-1_gamma_0_99.png
```

Observations

- As γ increases, the agent considers **future rewards more strongly**
- Lower γ (0.90):
  - Faster convergence
  - More short-term decisions
- Higher γ (0.99):
  - Slower convergence
  - More optimal long-term policy
- More complex grids (medium, hard) require **more iterations to converge**